

Actividad 1: Información de Países (REST Countries API)

Objetivo

Introducir el concepto de REST API, practicar peticiones GET, interpretar JSON y consumir una API pública con Postman y Python.

API gratuita (CON API Key)

Fase 1 – Exploración de la documentación

- Base URL: `https://restcountries.com/`
- Regístrate y obtener una API KEY

REST Countries

APIs Docs Log in Get an API key

Get an API key
Create a free account to access your API key.

FULL NAME
Gabriel Barrón Rodríguez

EMAIL
gbarron@utng.edu.mx

PASSWORD

CONFIRM PASSWORD

☐ I agree to the [Terms of Service](#) and [Privacy Policy](#).

¡Operación exitosa! CLOUDFLARE

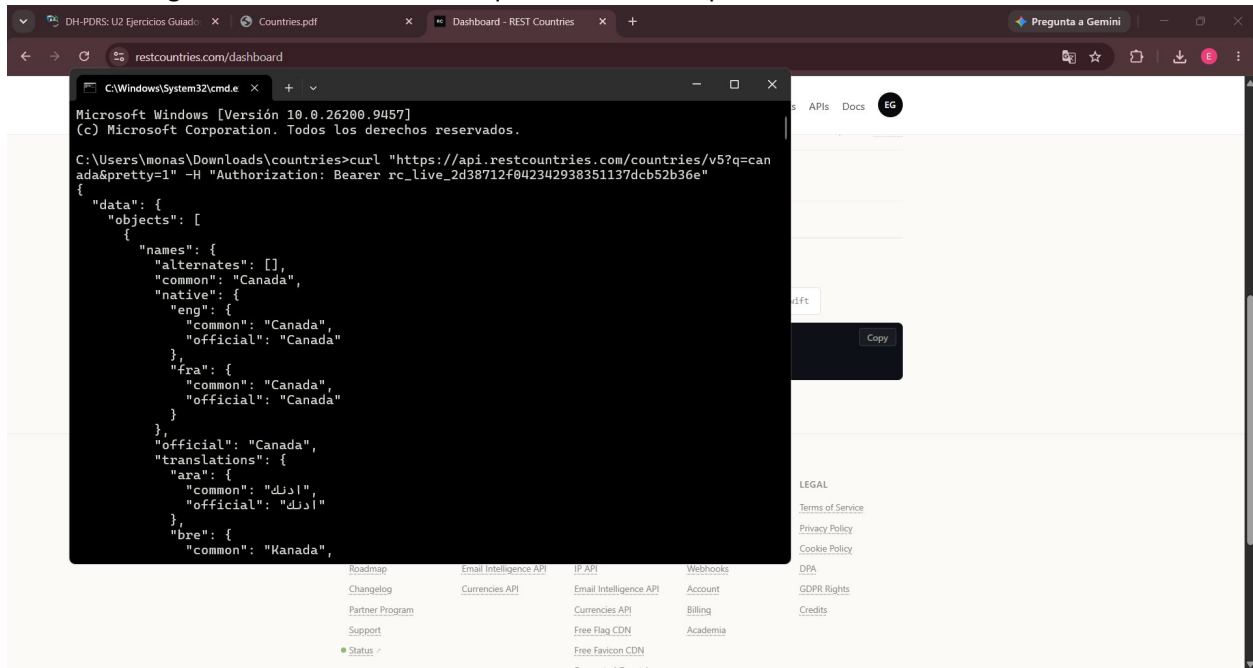
Create account

https://restcountries.com/sign-up#

- Guarda tu API Key de tal manera que no se te olvide
- Abre tu consola y teclea el siguiente comando

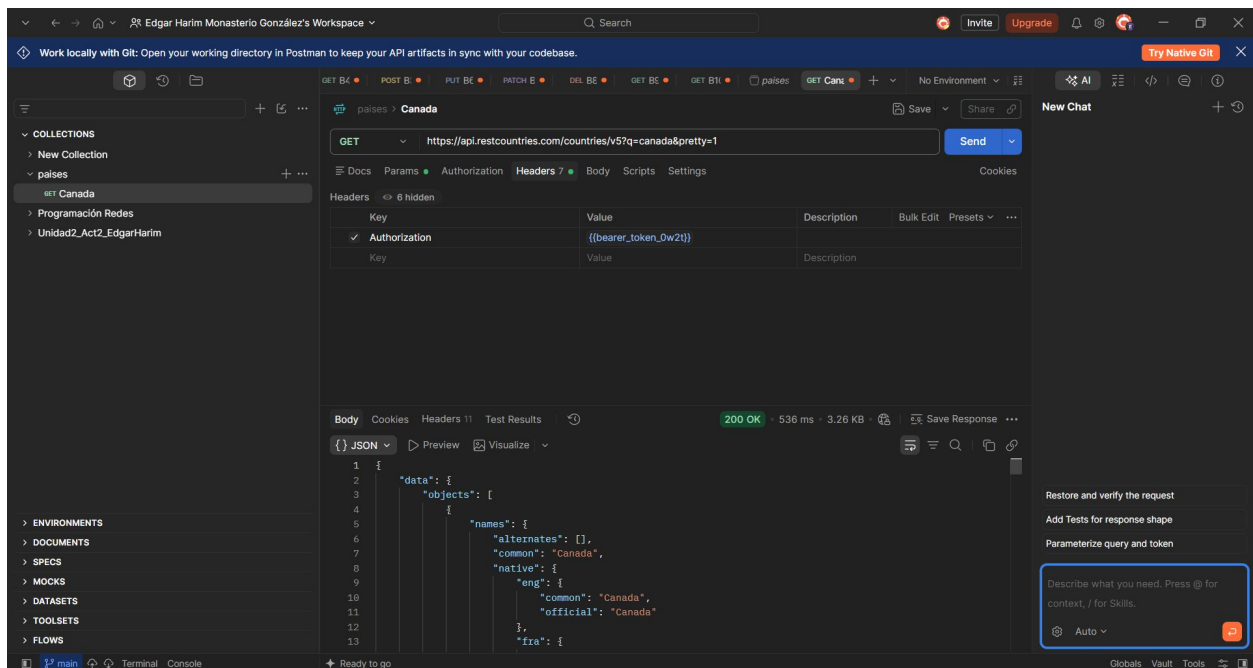
```
curl "https://api.restcountries.com/countries/v5?q=canada&pretty=1" -H "Authorization: XXXXXX"
```

- Pega tu resultado de manera que se visualice tu perfil



Postman

- Abre postman
- Crea una colección de API REST
- Agrega una nueva petición
- Selecciona el método **GET** y localiza la sección de HEADERS, agrega la clave **Autorization** y pega su contenido “Bearer API_KEY”



Código Python

```
import requests

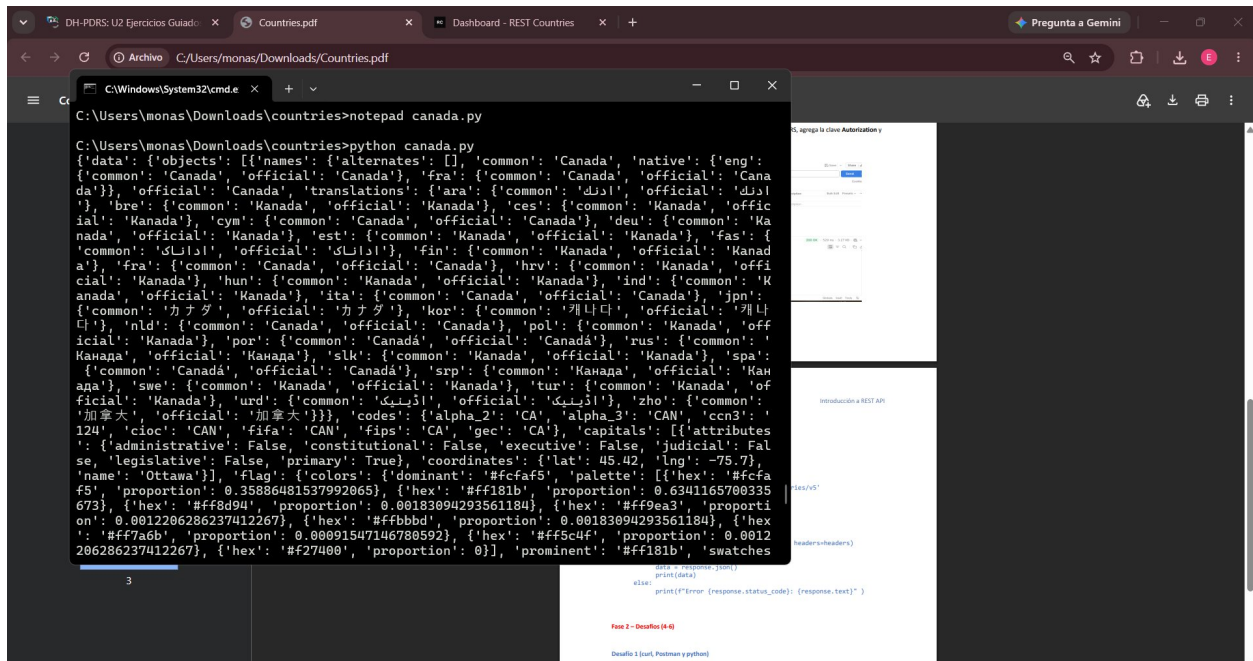
params = {
    "q": "canada",
    "pretty": "1"
}

url = 'https://api.restcountries.com/countries/v5'

headers = {
    "Authorization": 'Bearer
rc_live_5c958707a45848188d0e68bea868d00d'
}

response = requests.get(url, params=params, headers=headers)

if response.status_code == 200:
    data = response.json()
    print(data)
else:
    print(f"Error {response.status_code}: {response.text}")
```



```

C:\Windows\System32\cmd.exe
restcountries.com/dashboard

]]\d$}}, 'timezones': ['UTC-08:00', 'UTC-07:00', 'UTC-06:00', 'UTC-05:00', 'UTC-04:00',
'UTC-03:30'], 'tlds': ['.ca'], 'units': {'measurement_system': 'metric', 'temperature_sca
le': 'Celsius'}, 'uuid': '189581ed-44b0-47d7-9849-6b097401a7d6', '_match': [{'path': 'fla
g.description', 'value': 'The flag of Canada is composed of a red vertical band on the ho
ist and fly sides and a central white square that is twice the width of the vertical band
s. A large eleven-pointed red maple leaf is centered in the white square.'}, {'path': 'na
mes.native.eng.common', 'value': 'Canada'}, {'path': 'names.native.eng.official', 'value'
: 'Canada'}, {'path': 'names.native.fra.common', 'value': 'Canada'}, {'path': 'names.nati
ve.fra.official', 'value': 'Canada'}, {'path': 'names.common', 'value': 'Canada'}, {'path
': 'names.official', 'value': 'Canada'}, {'path': 'names.translations.cym.common', 'value'
: 'Canada'}, {'path': 'names.translations.cym.official', 'value': 'Canada'}, {'path': 'n
ames.translations.fra.common', 'value': 'Canada'}, {'path': 'names.translations.fra.offic
ial', 'value': 'Canada'}, {'path': 'names.translations.ita.common', 'value': 'Canada'}, {
'path': 'names.translations.ita.official', 'value': 'Canada'}, {'path': 'names.translatio
ns.nld.common', 'value': 'Canada'}, {'path': 'names.translations.nld.official', 'value':
'Canada'}], 'meta': {'lastUpdatedTimestamp': 1790182685}}, 'meta': {'total': 1, 'count'
: 1, 'limit': 25, 'offset': 0, 'more': False, 'request_id': 'b78c649b-3216-4889-869c-ba10
1de8647f', 'duration': 74}}

C:\Users\monas\Downloads\countries>python canada.py
Traceback (most recent call last):
  File "C:\Users\monas\Downloads\countries\canada.py", line 10, in <module>
    print(data["data"]["objects"][0]["name"]["common"])
KeyError: 'name'

C:\Users\monas\Downloads\countries>python canada.py
Canada

C:\Users\monas\Downloads\countries>

```

```

C:\Windows\System32\cmd.exe
restcountries.com/dashboard

ist and fly sides and a central white square that is twice the width of the vertical band
s. A large eleven-pointed red maple leaf is centered in the white square.'}, {'path': 'na
mes.native.eng.common', 'value': 'Canada'}, {'path': 'names.native.eng.official', 'value'
: 'Canada'}, {'path': 'names.native.fra.common', 'value': 'Canada'}, {'path': 'names.nati
ve.fra.official', 'value': 'Canada'}, {'path': 'names.common', 'value': 'Canada'}, {'path
': 'names.official', 'value': 'Canada'}, {'path': 'names.translations.cym.common', 'value'
: 'Canada'}, {'path': 'names.translations.cym.official', 'value': 'Canada'}, {'path': 'n
ames.translations.fra.common', 'value': 'Canada'}, {'path': 'names.translations.fra.offic
ial', 'value': 'Canada'}, {'path': 'names.translations.ita.common', 'value': 'Canada'}, {
'path': 'names.translations.ita.official', 'value': 'Canada'}, {'path': 'names.translatio
ns.nld.common', 'value': 'Canada'}, {'path': 'names.translations.nld.official', 'value':
'Canada'}], 'meta': {'lastUpdatedTimestamp': 1790182685}}, 'meta': {'total': 1, 'count'
: 1, 'limit': 25, 'offset': 0, 'more': False, 'request_id': 'b78c649b-3216-4889-869c-ba10
1de8647f', 'duration': 74}}

C:\Users\monas\Downloads\countries>python canada.py
Traceback (most recent call last):
  File "C:\Users\monas\Downloads\countries\canada.py", line 10, in <module>
    print(data["data"]["objects"][0][0]["name"]["common"])
KeyError: 'name'

C:\Users\monas\Downloads\countries>python canada.py
Canada

C:\Users\monas\Downloads\countries>python canada.py
Canada
Canada

C:\Users\monas\Downloads\countries>

```

Introducción a REST API

The screenshot shows a web browser with the REST Countries dashboard open. A terminal window is overlaid on the browser, displaying the following content:

```
ve.fra.official', 'value': 'Canada'}, {'path': 'names.common', 'value': 'Canada'}, {'path': 'names.official', 'value': 'Canada'}, {'path': 'names.translations.cym.common', 'value': 'Canada'}, {'path': 'names.translations.cym.official', 'value': 'Canada'}, {'path': 'names.translations.fra.common', 'value': 'Canada'}, {'path': 'names.translations.fra.official', 'value': 'Canada'}, {'path': 'names.translations.ita.common', 'value': 'Canada'}, {'path': 'names.translations.ita.official', 'value': 'Canada'}, {'path': 'names.translations.nld.common', 'value': 'Canada'}, {'path': 'names.translations.nld.official', 'value': 'Canada'}], 'meta': {'lastUpdatedTimestamp': 1790182685}}, 'meta': {'total': 1, 'count': 1, 'limit': 25, 'offset': 0, 'more': False, 'request_id': 'b78c649b-3216-4889-869c-ba101de8647f', 'duration': 74}}}
```

```
C:\Users\monas\Downloads\countries>python canada.py
Traceback (most recent call last):
  File "C:\Users\monas\Downloads\countries\canada.py", line 10, in <module>
    print(data["data"]["objects"][0]["name"]["common"])
KeyError: 'name'

C:\Users\monas\Downloads\countries>python canada.py
Canada

C:\Users\monas\Downloads\countries>python canada.py
Canada
Canada

C:\Users\monas\Downloads\countries>python canada.py
Nombre comun: Canada
Nombre oficial: Canada

C:\Users\monas\Downloads\countries>
```

Below the terminal, a code block shows the following Python code:

```
import requests
response = requests.get(
    'https://api.restcountries.com/countries/y5q?c=canada',
    headers={'Authorization': 'Bearer rc_live_2d38712f842342938351137dcb2b36e'})
```

The screenshot shows the REST Countries dashboard with the Canada entry selected. The terminal window displays the following content:

```
'Canada'}], '_meta': {'lastUpdatedTimestamp': 1790182685}}, 'meta': {'total': 1, 'count': 1, 'limit': 25, 'offset': 0, 'more': False, 'request_id': 'b78c649b-3216-4889-869c-ba101de8647f', 'duration': 74}}}
```

```
C:\Users\monas\Downloads\countries>python canada.py
Traceback (most recent call last):
  File "C:\Users\monas\Downloads\countries\canada.py", line 10, in <module>
    print(data["data"]["objects"][0]["name"]["common"])
KeyError: 'name'

C:\Users\monas\Downloads\countries>python canada.py
Canada

C:\Users\monas\Downloads\countries>python canada.py
Canada
Canada

C:\Users\monas\Downloads\countries>python canada.py
Nombre comun: Canada
Nombre oficial: Canada

C:\Users\monas\Downloads\countries>python canada.py
Nombre comun: Canada
Nombre oficial: Canada
Descripción: The flag of Canada is composed of a red vertical band on the hoist and fly sides and a central white square that is twice the width of the vertical bands. A large eleven-pointed red maple leaf is centered in the white square.

C:\Users\monas\Downloads\countries>
```

Below the terminal, an "EXAMPLE REQUEST" section shows a code block with the following Python code:

```
import requests
response = requests.get(
```